Quantum Computing - Week 1 Worksheet

Q1: Bra-Ket Notation

Express the following in Dirac notation:

1. The inner product of $|0\rangle$ and $|1\rangle$.
2. The outer product of $|0\rangle$ and $\langle 1|$.
3. The norm of the state $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$.

Q2: Matrix Operations

Compute the following:

1. $X|0\rangle$, where $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.
2. $XX|1\rangle$.
3. The result of applying the identity matrix I to $|1\rangle$.

Q3: Tensor Product

Compute the tensor product of the following states:

1. $|0\rangle \otimes |1\rangle$.
2. $|+\rangle \otimes |-\rangle$, where $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$.

Q4: Conceptual Questions

Answer the following:

1. What does it mean for two quantum states to be orthogonal?
2. Explain why the tensor product is necessary for multi-qubit systems.
3. How does the Pauli-X gate act on an arbitrary qubit state $|\psi\rangle = a|0\rangle + b|1\rangle$?